

# Why Predict Crop Yield in Kenya?

Crop yield varies across counties - making agricultural planning difficult.

- **Who is affected:** Farmers, agricultural officers, researchers, policymakers, planners
- **Why it matters:** Better yield data supports planning, food-security decisions, and resource allocation

📌 **Project Goal:** Predict the yield of 8 major food crops using historical agricultural data and satellite-derived environmental variables.

BEANS

COWPEAS

IRISH POTATOES

MAIZE

PIGEON PEAS

SORGHUM

SWEET POTATOES

WHEAT

2019–2023

Study Period

8

Food Crops

47

Counties

3

Data Sources

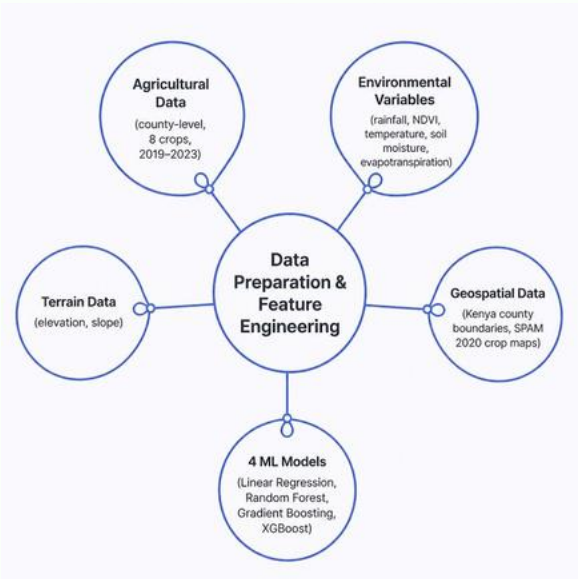
Agricultural, environmental, and geospatial

# From Agricultural Data to Yield Predictions

The project combines agricultural, environmental, and geospatial data to model crop yield at county level.



Agricultural, environmental, and geospatial data flow through feature engineering into four machine-learning models for comparison.



## Modeling Objective

**Target:** Crop Yield (t/ha)

### Linear Regression

Interpretable baseline

### Random Forest

Captures nonlinear patterns

### Gradient Boosting

Sequential error correction

### XGBoost

Optimized boosting

☐ **Mean Yield Baseline = 3.753 t/ha** — a benchmark for added predictive value.

# Random Forest Achieved the Best Overall Performance

0.859

MAE (t/ha) ↓  
Random Forest

1.623

RMSE (t/ha) ↓  
Random Forest

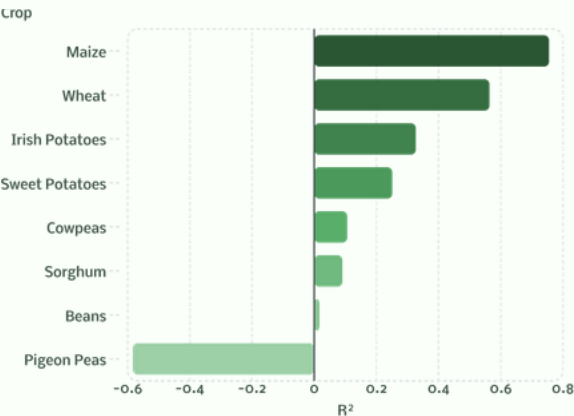
0.863

R<sup>2</sup> ↑  
Random Forest

Best overall performance among the four tested models.

| Model             | MAE (t/ha) ↓ | RMSE (t/ha) ↓ | R <sup>2</sup> ↑ | MAPE ↓ |
|-------------------|--------------|---------------|------------------|--------|
| Mean Baseline     | 3.566        | 4.391         | -0.004           | 334.4% |
| Linear Regression | 1.333        | 1.960         | 0.800            | 113.0% |
| Gradient Boosting | 0.917        | 1.719         | 0.846            | 50.8%  |
| XGBoost           | 0.887        | 1.677         | 0.854            | 40.3%  |
| Random Forest     | 0.859        | 1.623         | 0.863            | 41.5%  |

## Performance Varies Across Crops



## Crop-Level Evaluation

| Crop           | R <sup>2</sup> |
|----------------|----------------|
| Maize          | 0.7553         |
| Wheat          | 0.5636         |
| Irish Potatoes | 0.3268         |
| Sweet Potatoes | 0.2509         |
| Cowpeas        | 0.1060         |
| Sorghum        | 0.0905         |
| Beans          | 0.0168         |
| Pigeon Peas    | -0.5834        |

Overall accuracy alone does not tell the whole story. Maize is predicted most reliably; Pigeon Peas has negative R<sup>2</sup>, showing the model struggles to explain its yield variation.

# Where the Model Should -and Should Not - Be Trusted

## Limitations

- Cannot guarantee future harvests
- Depends on historical data quality and coverage
- Performance varies across crops
- Misses some agricultural factors



**The model is not equally reliable for all eight crops.**

Maize  $R^2 = 0.755$  vs. Pigeon Peas  $R^2 = -0.583$

## Responsible AI

### Who could be affected by errors?

- Farmers and agricultural planners
- County agricultural officers
- Communities underrepresented in training data

**Fairness criterion considered:** Individual Fairness — similar agricultural conditions should yield similar predictions, regardless of county identity.

## Responsible Deployment Cycle

**Monitor**  
Errors by crop and county

**Human Review**  
For high-impact decisions

**Investigate**  
Gaps in performance

**Retrain**  
When new data become available



**Use the model as decision support — not a replacement for agricultural expertise.**

# From Prediction to Responsible Agricultural Decision Support

## 🌿 Environmental Data Adds Value

Environmental and geospatial variables improve yield prediction alongside agricultural records.

## 🌲 Random Forest Led Overall

Best overall performance among the four tested ML models.

## 🏠 Performance Differs by Crop

Crop-level evaluation is essential; some crops perform much better than others.

## What Should Happen Next?

01

### Expand Training Data

Add more years of county yield observations

02

### Improve Representation

Reduce imbalance across crops and counties

03

### Test Crop-Specific Models

Study weaker crops and try specialized approaches

04

### Monitor After Deployment

Track errors and retrain as new data arrive

**"The goal is not simply to predict yield — it is to provide agricultural decision-makers with useful predictions while being transparent about where the model is reliable and where it is not."**